

Multiple Choice (10 marks)

1. What factors affect the rate of transpiration?
 - (a) Temperature, humidity, and air currents
 - (b) Plant species, root depth, and wind speed
 - (c) Soil fertility, irrigation methods, and cloud cover
 - (d) Water salinity, leaf color, and time of day
 - (e) Leaf surface area, stem length, and moon phase
2. What are the different types of point mutations?
 - (a) Frame shift mutations and nucleotide repeats
 - (b) Base substitutions and small insertions and deletions
 - (c) Base additions and large insertions
 - (d) Silent mutations and missense mutations
 - (e) Duplication mutations and triplet repeat expansions
3. Which is a true statement concerning genetic variation?
 - (a) It is created by the direct action of natural selection.
 - (b) It tends to be reduced by the processes involved when diploid organisms produce gametes.
 - (c) It must be present in a population before natural selection can act upon the population.
 - (d) It is always increased by the processes involved when diploid organisms produce gametes.
 - (e) It is not affected by the environmental changes.
4. Which group is composed entirely of individuals who maintained that species are fixed (i.e., unchanging)?
 - (a) Aristotle, Darwin, and Lamarck
 - (b) Aristotle, Lyell, and Darwin
 - (c) Aristotle, Linnaeus, and Cuvier
 - (d) Aristotle, Cuvier, and Lamarck
 - (e) Linnaeus, Darwin, and Lamarck
5. Compare the methods of obtaining oxygen in the earthworm, the frog, the fish and man.

- (a) Earthworms use gills, frogs use both gills and lungs, fish use lungs, and humans use skin.
- (b) Earthworms and humans use lungs, frogs and fish use gills.
- (c) All use gills for obtaining oxygen.
- (d) Earthworms and humans use gills, frogs use skin, and fish use lungs.
- (e) Earthworms use lungs, frogs use gills, fish use skin, and humans use gills.

True / False (10 marks)

Answer True or False

- 6. True or False: Search engine as a diagnostic tool in difficult immunological and allergologic cases: is Google useful?
- 7. True or False: Can increases in the cigarette tax rate be linked to cigarette retail prices?
- 8. True or False: Are head and neck specific quality of life measures necessary?
- 9. True or False: Did Chile's traffic law reform push police enforcement?
- 10. True or False: Is there a correlation between androgens and sexual desire in women?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the role of radioactive isotopes in biological research, particularly in tracing metabolic pathways within living cells, and analyze their impact on experimental observations.
- 12. Discuss the role of ethylene in the ripening of fruits, and evaluate the misconceptions surrounding the use of heat from kerosene stoves in this process during the early 1900s.

End of Paper